Day 6: /23 Subnetting and the MAC Address

Level 1, Month 1, Networking foundations

What I learned

- A MAC address is 48 bits = 6 bytes, written as 6 pairs of hexadecimal symbols.
- Each hexadecimal symbol is 4 bits, so one pair is 8 bits.
- Hexadecimal symbols: 0 to 9, then `A` = 10, `B` = 11, `C` = 12, `D` = 13, `E` = 14, `F` = 15.
- Converting a symbol to 4 bits with the table `8 4 2 1`:
  - `B` = 8 + 2 + 1 = `1011`
  - `5` = 4 + 1 = `0101`
  - `4` = `0100`
- I converted a whole MAC address (the one of my own adapter, not published here) pair by pair into 48 bits, and every conversion was correct.

Warm-up: 10.1.200.5/23

Item	Value
Bits d'hote	9
Bits in the third number	1 (24 - 23)
Taille du bloc	2
Adresse reseau	10.1.200.0
Premiere adresse	10.1.200.1
Broadcast	10.1.201.255
Derniere adresse	10.1.201.254
Hotes valides	510

Error log

Mistake	Correct answer	Why
Bits d'hote: 1	9	1 is the bits in the third number. Bits d'hote = 32 - 23, always from 32.
Hotes valides: 512	510	512 is all the addresses. Subtract 2.

Main lessons

- Two different subtractions: `32 - mask` for the bits d'hote, `24 - mask` (the boundary) for the bits in the number that carries the blocks. This confusion came back again today.
- Subtract 2 from the number of addresses to get valid hosts. I forgot it again, so I keep it on a card.
- Writing the sum next to each hexadecimal conversion (`8+2+1`) is the safest way to check it.

Anki

Cards for: the hexadecimal symbols A to F, `Bits d'hote = 32 - mask`, and `Hotes valides = 2^(bits d'hote) - 2`.

Next session

Look at an ARP packet and read the MAC address field in Wireshark, then a short subnetting review.